

Bug Hunters

It was realised a while ago that testing should be an integral part of the software lifecycle. The demand for software testers can only grow

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In October 1999, the \$125 million NASA Mars Climate Orbiter was believed to be lost in space as a result of a data conversion error: it was determined that the software used on the spacecraft had certain data in English units that should have been in metric units. Among other tasks, the orbiter was to serve as a communications relay for the Mars Polar Lander mission, which failed for unknown reasons in December 1999. Several investigating panels were convened to determine what allowed the error to go undetected, but as of 2000, NASA was still trying to figure out what caused the failure of the Mars Polar Lander.

Software bugs caused the bank accounts of 823 customers of a major US bank to be credited with \$924,844,208.32 each in May of 1996. The American Bankers Association claimed it was the largest such error in banking history.

Moral of the story: in a world increasingly dependant on software, the consequences of bugs in programs can be majorly disruptive—or even catastrophic.

No Monkeying Around

The software lifecycle begins when an application is first conceived and ends when it is no longer in use. Among the various stages, test planning, testing, and re-testing are therefore vital. It is this function that the Software Tester or Quality Analyst (QA) fulfils.

Software Testing has come a long way from the days of “monkey testing.” According to Abhay Joshi, chief engineering officer of Disha Technologies, a Pune-based independent testing firm, “Testing used to be an afterthought. In the early days of mainframes, and later on with the PC, it was thought that if a monkey entered all sorts of random input sequences, hidden bugs in the software would be found. As if real users would behave like monkeys instead of doing their routine work!”

Jobs For The Asking

For those considering careers in testing, the Indian scene looks optimistic. Apart from the typical opportunities in development firms, testing is an area where considerable outsourcing action is projected. Also, QA is often the starting point for many clients, as there is less apprehension about Intellectual Property violation. According to market intelligence firm IDC, the global market for software testing is expected to touch \$13 billion by 2007. Of this, \$4 to \$5 billion is the offshore share. It is this part of the pie that is expected to increase the Indian market size from \$800 million at present to \$2 billion by 2007.

Pankaj Agarwal, CEO, Sopra Group (India), a subsidiary of the European consulting, systems integration and application outsourcing company Sopra, Newell and Budge, speaks about the thought behind his company's plan to recruit 800 test engineers in the next 18 months at their Noida facility. “Testing services as part of the overall revenues have gone up considerably. Also,

